Level 3 - Hydrotest with gasketed connection.

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1. Purpose

This document defines the guidelines to for the analysis of pump, OH or BB, in hydrotest condition with gasket. This procedure completes and integrates the *SOP-527 Gasket selection and verification* and *SOP-530 Level 3 Hydrotest with contact and bolt verifications*.

2. Abbreviations

BB	between bearing pump
DOF	Degree Of Freedom
OH	overhung pump
D_c	consistent tangent to pressure/closure curve [MPa/mm]
E	Young's Modulus [MPa]
G	Shear Modulus [MPa]
h	initial thickness of the gasket [mm]
Δc	total compression [mm]
Δc^e	total elastic compression [mm]
Δc^p	total plastic compression [mm]
ε	strain [-]
σ	stress [MPa]
ν	Poisson coefficient [-]

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3. Scope

This procedure is applicable to all pressure components that:

- sustain an internal pressure;
- are connected with bolts or studs;
- have a gasket between flanged connections;
- are required to be in accordance with international standards about leakage between bolted planes.
- the procedure is applicable to the plant of Nova Milanese and the plant of La Spezia.

Bolted joint, contact and bolt verification are described in *SOP-530 - ENG -Level 3 - Hydrotest with contact and bolt verifications.* This document expands and completes what was explained in that document.

4. Basic knowledge

Gaskets are used to seal the metal parts of the pump to prevent fluid from escaping. They are complex (often multi-layer) components, usually rather thin and typically made of several different materials of varying thickness. The gaskets are carefully designed to have a specific behaviour in the thickness direction. This is to ensure that the joints remain sealed when the metal parts are loaded by thermal or mechanical loads. The through-thickness behaviour, usually expressed as a relation between the pressure on the gasket and the closure distance of the gasket, is highly nonlinear, often involves large plastic deformations, and is difficult to capture with a standard material model in a FEM study. The alternative of modelling the gasket in detail by taking every individual material into account in the finite element model of the engine is not feasible: it requires a lot of elements which makes the model unacceptably large. Also, determining the material properties of the individual materials might be cumbersome.

For these reasons an alternative approach is developed.

The gasket material model addresses these problems by allowing gaskets to be modelled with only one element through the thickness, while the experimentally or analytically determined complex pressure-closure relationship in that direction can be used directly as input for the material model.

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5. Constitutive model

The behaviour in the thickness direction, the transverse shear behaviour, and the membrane behaviour are fully uncoupled in the gasket material model. In subsequent sections, these three deformation modes are discussed.

5.1. Local coordinate system

The material model is most conveniently described in terms of a local coordinate system in the integration points of the element (see Figure 1). For three-dimensional elements, the first and second directions of the coordinate system are tangential to the mid-surface of the element at the integration point. The third direction is the thickness direction of the gasket and is perpendicular to the mid-surface. For two-dimensional elements, the first direction of the coordinate system is the direction of the mid-surface at the integration point, the second direction is the thickness direction of the gasket and is perpendicular to the mid-surface, and the third direction coincides with the global three direction.

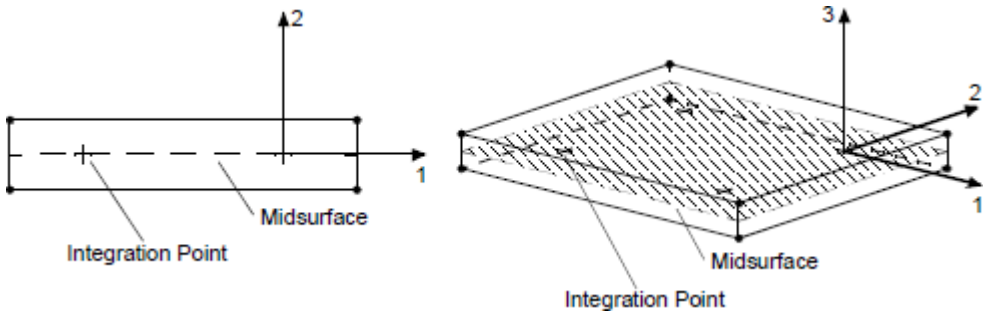

Figure 1 Integration points and local coordinate system.

In a total Lagrange formulation, the orientation of the local coordinate system is determined in the undeformed configuration and is fixed. In an updated Lagrange formulation, the orientation is determined in the current configuration and is updated during the analysis

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5.2. Thickness Direction - Compression

In the thickness direction, the material exhibits the typical gasket behaviour in compression, as depicted in Figure 2. After an initial nonlinear elastic response (section AB), the gasket starts to yield if the pressure p on the gasket exceeds the initial yield pressure p_{y0}. Upon further loading, plastic deformation increases, accompanied by (possibly nonlinear) hardening, until the gasket is fully compressed (section BD). Unloading occurs in this stage along nonlinear elastic paths (section FG, for example). When the gasket is fully compressed, loading and unloading occurs along a new nonlinear elastic path (section CDE), while retaining the permanent deformation built up during compression. No additional plastic deformation is developed once the gasket is fully compressed.

The loading and unloading paths of the gasket are usually established experimentally by compressing the gasket, unloading it again, and repeating this cycle a number of times for increasing pressures. The resulting pressure-closure data can be used as input for the material model. If the pressure also varies with temperature, then pressure-closure data at different temperatures can be provided. The user must supply the loading path and may specify up to ten unloading paths. In addition, the initial yield pressure p_{y0}. must be given. The initial yield pressure can also be varied with temperature and spatial coordinates. The loading path should consist of both the elastic part of the loading path and the hardening part, if present. If no unloading paths are supplied or if the yield pressure is not reached by the loading path, the gasket is assumed to be elastic. In that case, loading and unloading occurs along the loading path.

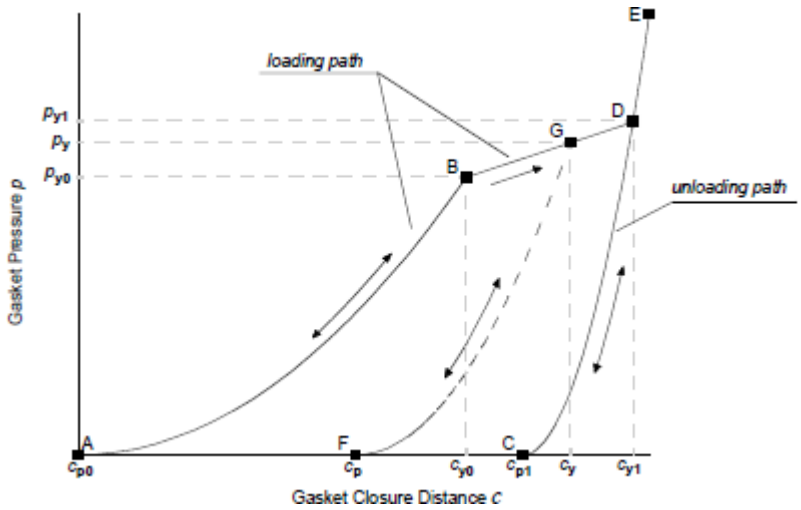

Figure 2 Gasket pressure vs closure.

The loading and unloading paths must be defined using table and must relate the pressure on the gasket to the gasket closure. Optionally, at each closure value, the loading and unloading paths can also be defined as functions of temperature and spatial coordinates using multi-variate table. The

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unloading paths specify the elastic unloading of the gasket at different amounts of plastic deformation; the closure at zero pressure is taken as the plastic closure on the unloading path. If unloading occurs at an amount of plastic deformation for which no path has been specified, the unloading path is constructed automatically by linear interpolation between the two nearest user supplied paths. The unloading path, supplied by the user, with the largest amount of plastic deformation is taken as the elastic path at full compression of the gasket. a total Lagrange formulation, the orientation of the local coordinate system is determined in the undeformed configuration and is fixed.

For example, in Figure 2, the loading path is given by the sections AB (elastic part) and BD (hardening part), and the initial yield pressure is the pressure at point B. The (single) unloading path is curve CDE. The latter is also the elastic path at full compression of the gasket. The amount of plastic closure on the unloading path is c_{p1}. The dashed curve FG is the unloading path at a certain plastic closure c_p that is constructed by interpolation from the elastic part of the loading path (section AB) and the unloading path CD.

The compressive behaviour in the thickness direction is implemented by decomposing the incremental gasket closure into an elastic and a plastic part:

$$\Delta = \Delta c^e + \Delta c^p$$

$$\Delta c = total\ compression$$

$$\Delta c^e = elastic\ compression$$

$$\Delta c^p = plastic\ compression$$

Of these two parts, only the elastic part contributes to the pressure. The constitutive equation is given by the following equation:

$$\Delta p = D_c \Delta c^e = D_c\left(\Delta c - \Delta c^p\right)$$

$$D_c = consistent\ tangent\ to\ pressure/closure\ curve$$

Plastic deformation develops when the pressure p equals the current yield pressure p_{y0}. The latter is a function of the amount of plastic deformation developed so far and is given by the hardening part of the loading path (section BD in Figure 2).

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5.3. Initial Gap

The thickness of a gasket can vary considerably throughout the sealing region. Since the gasket is modelled with only one element through the thickness, this can lead to meshing difficulties at the boundaries between thick regions and thin regions. The initial gap parameter can be used to solve this. The parameter basically shifts the loading and unloading curves in the positive closure direction. As long as the closure distance of the gasket elements is smaller than the initial gap, no pressure is built up in the gasket. The sealing region can thus be modelled as a flat sheet of uniform thickness and the initial gap parameter can be set for those regions where the gasket is actually thinner than the elements of the finite element mesh used to model it. The initial gap can optionally be varied as a function of spatial coordinates by using a table.

5.4. Thickness Direction - Tension

The tensile behaviour of the gasket in the thickness direction is linear elastic and is governed by a tensile modulus D_t. The latter is defined as a pressure per unit closure distance (that is, length). D_t can optionally be varied as a function of temperature and spatial coordinates by using a multi-variate table.

5.5. Transverse Shear and Membrane Behaviour

The transverse shear is defined in the 2-3 and 3-1 planes of the local coordinate system (for three-dimensional elements) or the 1-2 plane (for two-dimensional elements). It is linear elastic and characterized by a transverse shear modulus G_{t1}.

In three-dimensional elements only, orthotropic transverse shear behaviour may alternatively be characterized by the first transverse shear modulus in the 2-3 plane G_{t1} and by the second transverse shear modulus in the 3-1 plane G_{t2}. The shear moduli can optionally be varied as a function of temperature and spatial coordinates by using a multi-variate table.

The membrane behaviour is defined in the local 1-2 plane (for three-dimensional elements) or the local 3-1 plane (for two dimensional elements) is linear elastic and may be isotropic or orthotropic. Young's modulus E_m and Poisson's ratio ν_m that govern the membrane behaviour are taken from an existing material that must be defined.

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5.1. Thermal Expansion

The thermal expansion of the gasket material is isotropic, and the thermal expansion coefficient is taken from the material that also describes the membrane behaviour.

5.1. Constitutive Equations

As mentioned above, the behaviour in the thickness direction of the gasket is formulated as a relation between the pressure $p$on the gasket and the gasket closure distance c. In order to formulate the constitutive equations of the gasket material, this relation must first be written in terms of stresses and strains. This depends heavily on the stress and strain tensor employed in the analysis. For small strain analyses, for example, the engineering stress and strain are used. In that case, the incremental gasket closure and the incremental pressure are related to the incremental strain and the incremental stress by:

$$\Delta c = - h\Delta\varepsilon$$

$$\Delta p = - \Delta\sigma$$

$$h = initial\ thickness\ of\ the\ gasket$$

For example, the resulting constitutive equation for three-dimensional isotropic elements, expressed in the local coordinate system of the integration points, is

$$\begin{bmatrix} \Delta\sigma_{11} \\ \Delta\sigma_{22} \\ \Delta\sigma_{33} \\ \Delta\sigma_{12} \\ \Delta\sigma_{23} \\ \Delta\sigma_{31} \end{bmatrix} = \begin{bmatrix} \frac{E}{1-v^2} & \frac{vE}{1-v^2} & 0 & & & \\ \frac{vE}{1-v^2} & \frac{E}{1-v^2} & 0 & & 0 & \\ 0 & 0 & hD_c & & & \\ & & & \frac{E}{2(1+v)} & 0 & 0 \\ & 0 & & 0 & G_t & 0 \\ & & & 0 & 0 & G_t \end{bmatrix} \begin{bmatrix} \Delta\varepsilon_{11} \\ \Delta\varepsilon_{22} \\ \Delta\varepsilon_{33} - \Delta\epsilon_{33}^{p} \\ \Delta\gamma_{12} \\ \Delta\gamma_{23} \\ \Delta\gamma_{31} \end{bmatrix}$$

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6. Gasket curves

Gasket curves can be evaluated following the procedure described in *SOP-527 Gasket selection and verification.*

7. Bolt/stud and contact verification

FEA model of the gasket gives the contact pressure distribution on all elements. The allowable value necessary to guarantee level of leakage in accordance with international standard can be evaluated using *SOP-527 Gasket selection and verification.*

Bolt stress must be in accordance with *SOP-530 Level 3 Hydrotest with contact and bolt verifications.*

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8. References

- *Marc 2024.2 Volume A: Theory and User Information.*
- *G.10.00.21 - Guidelines for static analysis complexity level definition*
- *SOP-530 - Level 3 Hydrotest with contact and bolt verifications.*
- *SOP-527 - Gasket selection and verification.*

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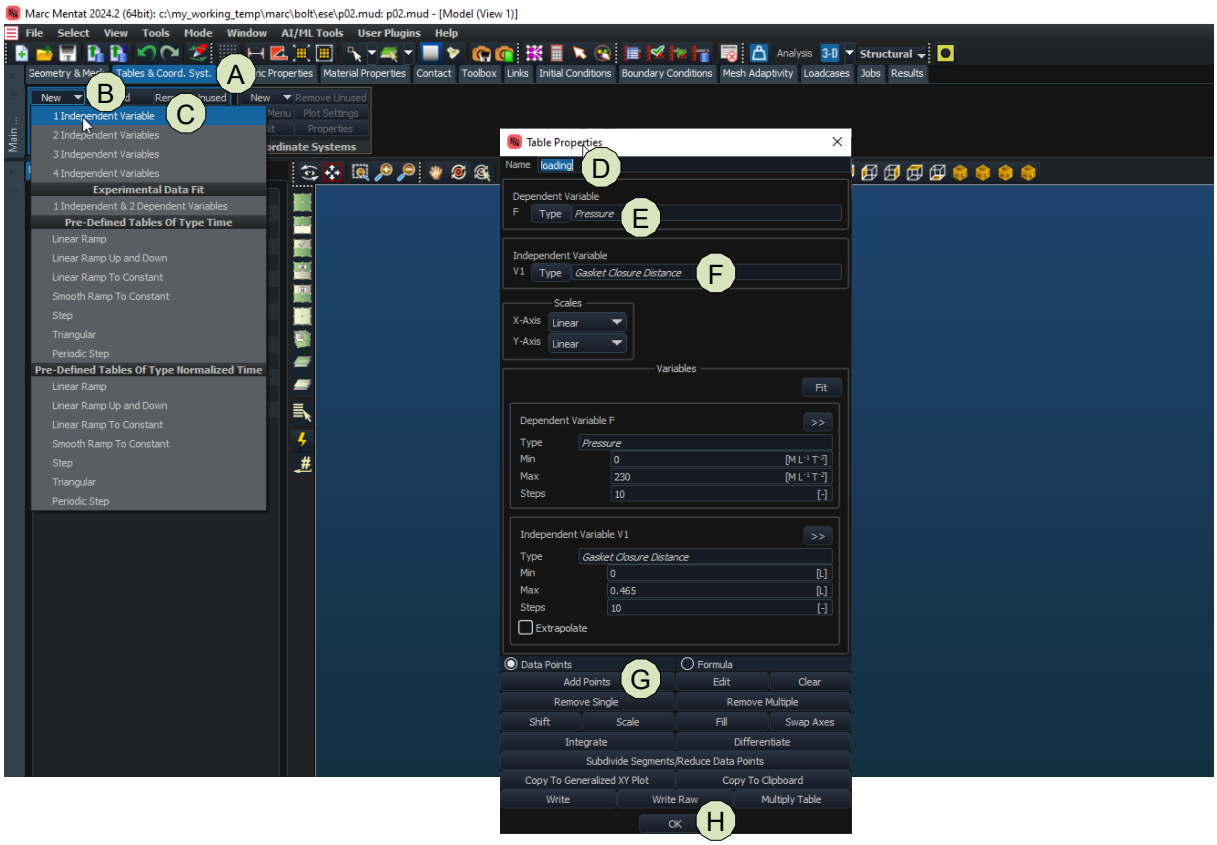
9. Annex A: Marc procedure

This simplified procedure is based on Marc 2024.2. A detailed description of these commands is available in Marc documentation.

9.1. STEP 1: create table "loading".

Create a table for gasket loading:

A. Table & Coordinate Syst.
B. New
C. 1 Independent Variable
D. Name → "loading"
E. Dependent Variable Type → Pressure
F. Independent Variable Type → Gasket Closure Distance
G. Add Points
H. OK



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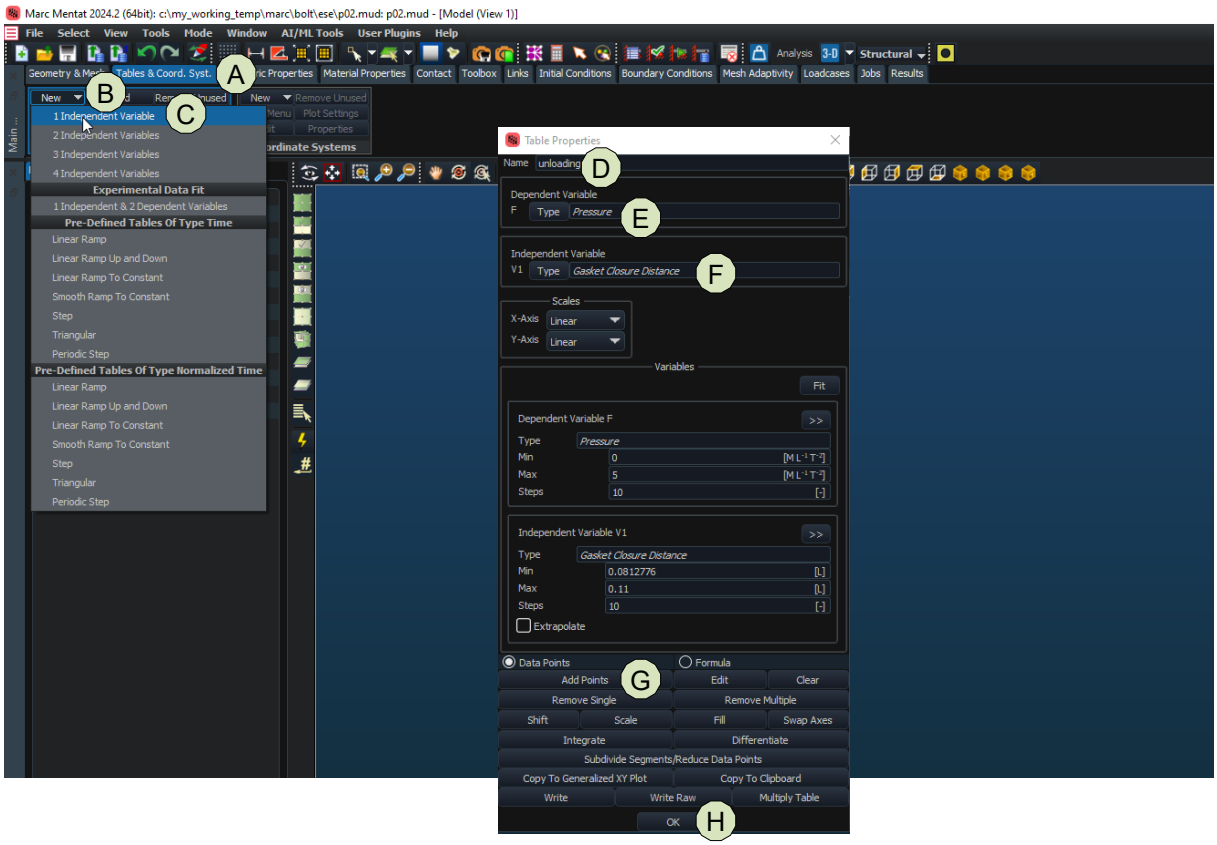
9.2. STEP 2: create table "unloading1".

Create a table for unloading1 (repeat for the n unloading curves available):

A. Table & Coordinate Syst.
B. New
C. 1 Independent Variable
D. Name → "unloading1"
E. Dependent Variable Type → Pressure
F. Independent Variable Type → Gasket Closure Distance
G. Add Points
H. OK



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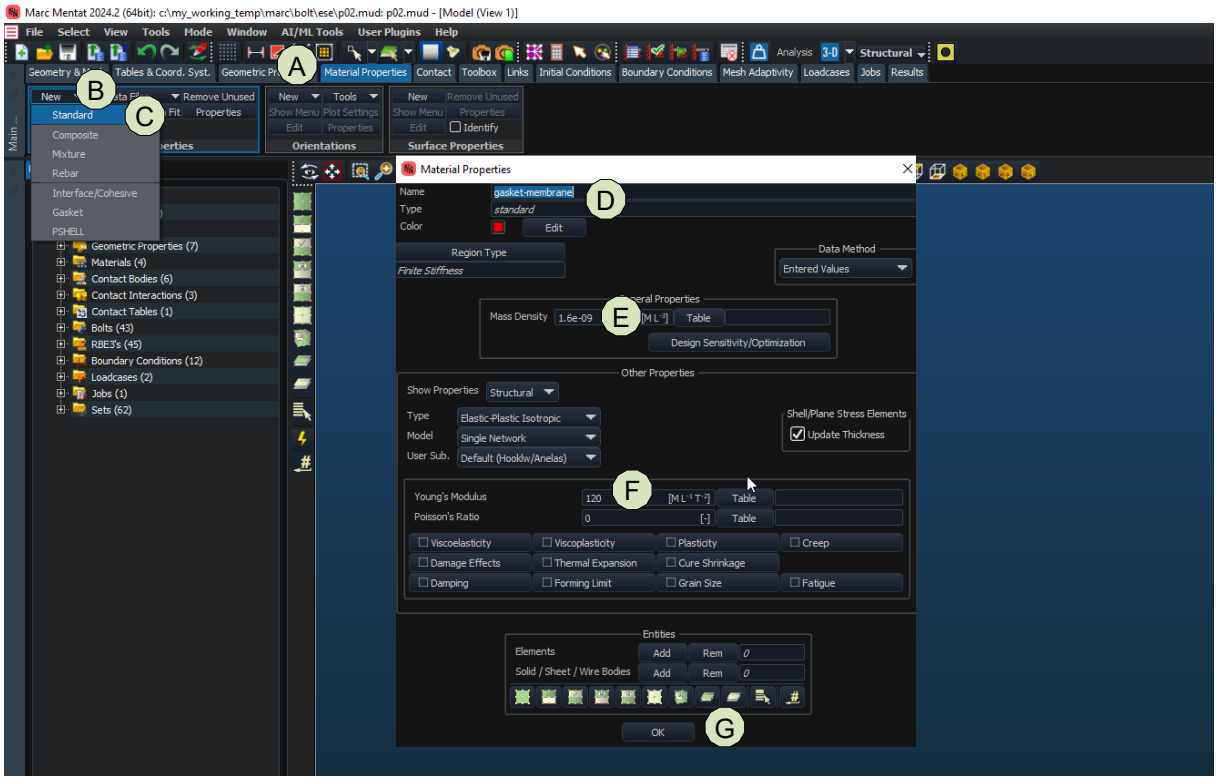
9.3. STEP 3: create membrane behaviour material.

Create membrane gasket material:

A. Materials Properties
B. New
C. Standard
D. Define the name
E. Define density
F. Define Young's modulus
G. OK



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9.4. STEP 4: create gasket material.

Create gasket material:

A. Materials Properties
B. New
C. Gasket
D. Define the name
E. Select membrane behaviour
F. Define Yield Pressure
G. Define Tensile Modulus
H. Define Initial Gap
I. Select loading curve
J. Select unloading curves (max10)
K. Define Transverse Shear Modulus
L. Add elements
M. OK

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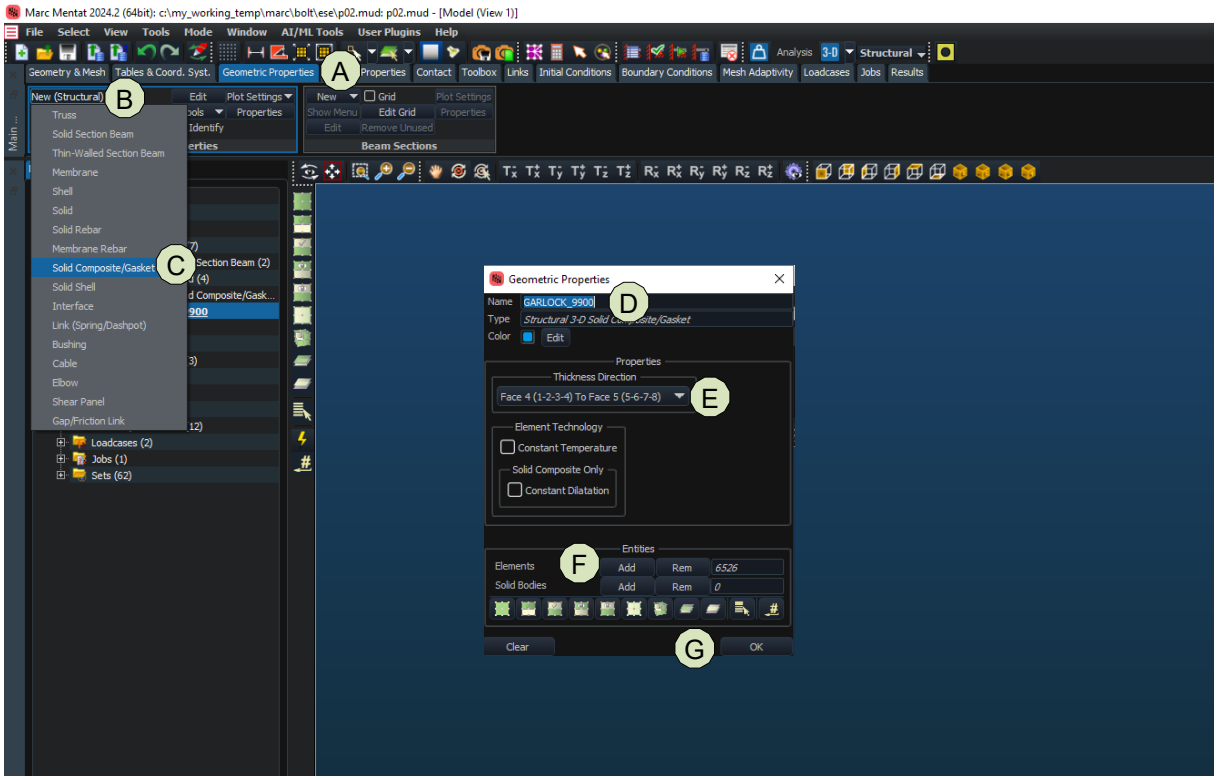
9.5. STEP 5: create gasket geometric properties.

Create gasket geometric properties:

A. Geometric Properties.
B. New (Structural)
C. Solid Composite/Gasket
D. Define Name
E. Select Thickness Direction
F. Add elements
G. OK



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